

2026 Changelog of Computing Computing Bootcamp

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Change Log

The following is a list of changes for how this course has been changed since last year.

2026

- The course now consists of 5 modules instead of 10 modules.

Module 1

- Encapsulates Modules 1-3 of the 2025 bootcamp
- Including the addition of `devtools` and `pak` packages for installing packages.
- Including the mention of Quarto and `qmd` files
- Referencing “R for Data Science Books (2e)” book
- replacing Google style Guide with Tidyverse style guide
- Removing *when* solutions will be posted.
 - Unsure about if there will be time for the class to work on exercises during lecture or not
- Not including talking points about LLMs- I plan on talking about it in passing

Exercises

- Referencing the `RKaggle` package to download datasets directly from Kaggle without having to manually download the csv

Module 2

- Encapsulates Modules 4-5 of the 2025 bootcamp
- Remove the discussion of Parametric vs Nonparameteric Models & Frequentist vs Bayesian Discussion
- Remove Method of Moments Discussion
- Adding mention of `method="BFGS"` in comments for code for using the `optim` function
- Removing discussion of the EM algorithm (Coursework will cover that)
 - Using the `optim()` function is useful
 - Newton Raphson is helpful to know as well.
 - Remove Delta Method

Module 3

- Encapsulates Module 6 of the 2025 bootcamp
- Remove Wald and Score Tests.
- Remove Test Equivalence
- Adding explanation of the Fox (1997) visual comparing the LR, Wald and Score tests.

Module 4

- Encapsulates Modules 7-8 of the 2025 bootcamp
- Removing Mean, Variance and MGF calculation of the mean of EFs
 - Not relevant for applications
 - It will be covered in coursework.
- Removing Discussion of IRLS
 - It will be covered in coursework.

Module 5

- Encapsulates Modules 9-10 of the 2025 bootcamp
- Noting that `mclapply` does not work on Windows Machines
- Removing Discussion on HPC (not relevant for many Msc students and first year PhD students)
- Removing Bootstrap